

Economic order/production quantity (EOQ/EPQ) models with product recovery: A review of mathematical modeling (1967–2022)

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ABSTRACT

This paper reviews the mathematics of the economic order/production quantity (EOQ/EPQ)-based models in the reverse logistics (RL) literature. It starts with the seminal work of Schrady (A deterministic inventory model for repairable items. Naval Research Logistics Quarterly, 14(3), 1967, 391–398) up until December 31, 2022. It provides researchers with a road map of how the mathematics has evolved since the work of Schrady and saves them time and effort by avoiding the need to review the entire RL literature. As it unifies the notations for all reviewed models, it also provides some research pointers for promising topics to be pursued.

Introduction

Since the Industrial Revolution and until a few decades ago, humans have operated in a linear economy (extract, produce, use, and dispose) on the premise that the earth's natural resources will never dry out. We have since come to realize that these resources are finite and that the rapid increase in the world's population and lifestyle changes have resulted in many problems and environmental legacies [1]. The most pressing of these include how to (1) reduce greenhouse gas (GHG) emissions that result in and continue to contribute to climate change, (2) sustain production without extracting virgin material, and (3) reduce waste disposed of in landfill sites or incinerators that pollutes soil and freshwater tables.

The concept of circular economy (CE) is not new. It was initially termed “cyclic ecological systems” and took a boost in the 1990s along with reverse logistics (RL) [2] due to the rise in societal and governmental environmental concerns. RL and CE share similarities. Both deal with repair, refurbishing, remanufacturing, recycling, and disposal activities. CE, however, is broader as it is more holistic in covering concepts relating to economic, environmental, and social sustainability [3,4]. Specifically, CE is a regenerative system, whose objective is to minimize resource input and waste, GHG emissions, and energy leakage by slowing, closing, and narrowing material and energy loops [4]. CE can be achieved through extended design, maintenance, repair, reuse, remanufacturing, refurbishing, and recycling [4]. CE incorporates features and contributions from an array of concepts that share the idea of closed loops [4,5], a concept adopted in closed-loop supply chains (CLSC = forward + reverse logistics). Inventory can take many forms. It is tangible, i.e.,

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Nomenclature: Important note: Only the parameters and variables common to more than one equation/model are listed here. There are instances where a parameter is specific to one model only, that has been defined in the text where appropriate.

Latin

c_d	unit-cost (value) of a non-serviceable stock
c_p	unit production (procurement) cost (\$/unit)
c_r	unit repair cost (\$/unit)
c_s	unit-cost (value) of a serviceable stock
c_w	unit waste disposal cost (\$/unit)
D	demand rate (units/unit-time)
D_p, D_r	demand rate for new and repaired/remanufacture/recovered items (units/unit-time)
h_d	holding cost for returned recoverable (non-serviceable) item (\$/unit/unit-time)
h_p	holding costs for production stock (\$/unit/unit-time)
h_r	holding costs for repair/remanufacture (non-serviceable) stock (\$/unit/unit-time)
h_s	holding costs for serviceable (new and repaired/remanufactured) stock (\$/unit/unit-time)
K_p	fixed setup cost of production run (\$/run)
K_r	fixed setup cost of repair run (\$/run)
m	number of repair/remanufacture setups (or batches) in one cycle of length T
m^*	fixed number of recovery batches in one cycle
n	number of production setups (or batches) in one cycle of length T
P	production rate (units/unit-time)
Q	lot size quantity (new + repair)
Q_p	production (procurement) quantity (units)
Q_r	repair quantity (units)
q	quality level of returned items, $0 < q < 1$
R	repair (recovery) rate (units/unit-time)
T	cycle time
T_p, T_r	lengths of production and remanufacturing intervals

Greek

α	fraction of items scrapped (% of demand), $\alpha = 1 - \beta$
β	fraction of items recovered (repaired) (% of demand)
$\beta_p(\beta_r)$	fraction of items recovered;% of demand for primary (secondary) market for produced (recovered)
γ	collection rate of available returns ($0 < \gamma < 1$)
$\gamma_p(\gamma_r)$	collection rate of available returns of newly produced (previously remanufactured) items

something to be mined, converted, created, transported, and sold [6]. Our lives depend on inventory [6], and, therefore, companies, and subsequently CLSCs, are affected by it, giving rise to the need for sustainable inventory and logistics management [1].

The norm in supply chains has been to move products with short lives faster to meet continual changes in customers' demand, subsequently, resulting in solid waste and depletion of natural resources at alarming rates [1]. The notion that the world's resources are finite, societal pressures, and governmental legislation pushed manufacturers to initiate product recovery programs to extend the useable lives of products. Recovery programs drive RL where used products are collected from customers or designated points, transported, stocked, inspected, and dismantled into components for reuse, remanufactured/repared, recycled, or disposed. Some researchers (e.g., Richter [7,8], Teunter [9,10]) captured this opportunity in the 1990s to develop models that help us understand how to manage inventories in systems with circular flows.

This paper aims to provide a concise review of the mathematical modeling of inventory models for RL. The scope focuses on those works that focus on producing, selling, collecting, and recovering (reusing, repairing, refurbishing, remanufacturing, and recycling). Of course, reviewing the mathematics of all those works will be impossible for one paper to consider. So, this paper only focuses on those works that used the economic order/production quantity (EOQ/EPQ) model as it has been the edifice and the first scientific approach reported in inventory theory and management and continues to be celebrated by many researchers [11,12]. Specifically, our research objective is to review the works stemming from a seminal work of Schrady [13], "A deterministic inventory model for reparable items", to find out how the mathematics of this field has evolved, what was covered and what is still to be covered, and to unify the notations of all reviewed models. A brief literature review to help frame the current paper in the right context is next. The sub-sections after that will focus on product recovery, inventory models with product recovery, the evolution and reviews of EOQ/EPQ models with product recovery, and our search of the literature.

Product recovery

Extending the useful life of equipment through repair or overhaul was a concept initiated by the U.S. military in the 1960s [13]. The rationale is to extend the useful life of equipment or products that are expensive to replace. The product recovery concept generally encompasses “the handling of all used and discarded products, components, and materials” [14], which initiates RL in a supply chain. The recovery process may vary, depending on the condition of used products, among others, which is seen in the extent of the return flow in the supply chain (Fig. 1). In this paper, the term “recovery” generally refers to remanufacturing; however, it could also refer to any action that brings a product or its components back to life through reuse, repair, refurbishing/reconditioning, and/or recycling [15]. Repair, for example, is the correction of specific faults in a product. Reconditioning expands the work by rebuilding major components; however, it does not return to its original like-new condition. Remanufacturing refers to disassembling a product, following manufacturer specifications, then rebuilding a product from used and new components, and giving it warranties equal to those of equivalent new products. Fig. 1 illustrates the processes that fall under the umbrella of recovery.

Inventory models with product recovery

Managing the inventory of purchasing or ordering new equipment and collecting those that need recovery is a complex system to model mathematically and, if so, to solve [13,16]. The decisions to consider include how often to review the stock status and when and how many returned items to recover and new ones to order at a time [14]. The literature classifies mathematical models into continuous review, periodic review, queueing-based, and other (heuristics, simulation, etc.) policies with single- or multi-echelon [17], where each echelon or tier is responsible for one or more activities in the supply chain (Fig. 1).

Continuous review inventory models have their inventory levels known. They are of two types: one-for-one and batch (or EOQ/EPQ) replenishment. With a one-for-one policy, any time a product fails in use, it is immediately sent to the recovery facility [18], where the failed part is replaced from inventory or when the ordered part arrives. A one-for-one policy is often used for expensive and low-demand items when the order setup cost is low compared to the shortage and holding costs and when the product demand and returns correlate perfectly (e.g., spare parts and repair management) [19]. It does not usually fit with remanufacturing when the objective is determining when and how much to remanufacture, dispose, and produce to minimize the cost function. EOQ/EPQ policies suit inventory situations with high demand rates and/or ordering costs. In a *periodic review* system, the inventory level is known only at discrete points when physical stock-taking occurs. Today’s inventory systems typically follow continuous reviews when parts tracking is automated using barcodes, etc. In such systems, and unlike periodic review systems, reorder decisions are made at any time rather than at predetermined times at the start of periods. However, it is common in the literature to approximate periodic models assuming continuous review instead. A queueing-based approach is appropriate when recovery capacity is limited (e.g., continuous review model) and other policies are for more complex situations [20,21].

Inventory models can also be classified as deterministic and stochastic [22,23]. *Deterministic* models have all parameters, including demand and returns, known and constant. Real-life systems are often *stochastic* when items fail at different random times and must be repaired or replaced; the decision process is how much (i.e., the level of) inventory to stock either initially, periodically, or continuously. Covering the mathematical models following the above classification of inventory systems would be excessive for one review paper. The focus here will be on deterministic models, i.e., those that follow the line of Schrady [13] and use the EOQ/EPQ modeling approaches. Although EOQ/EPQ assumptions are usually unmet in practice, EOQ/EPQ models remain popular among researchers and practitioners due to their unsophisticated mathematics (e.g., [14]). Like other operations management researchers, the authors of this article believe that the EOQ/EPQ will continue to be used as a fundamental model on which building more complex models is based. This emphasis will further justify the focus on such models in this article.

Evolution and reviews of EOQ/EPQ models with product recovery

The EOQ/EPQ model is the first scientific inventory management model [11,24]. Although it is a simple model and does not always

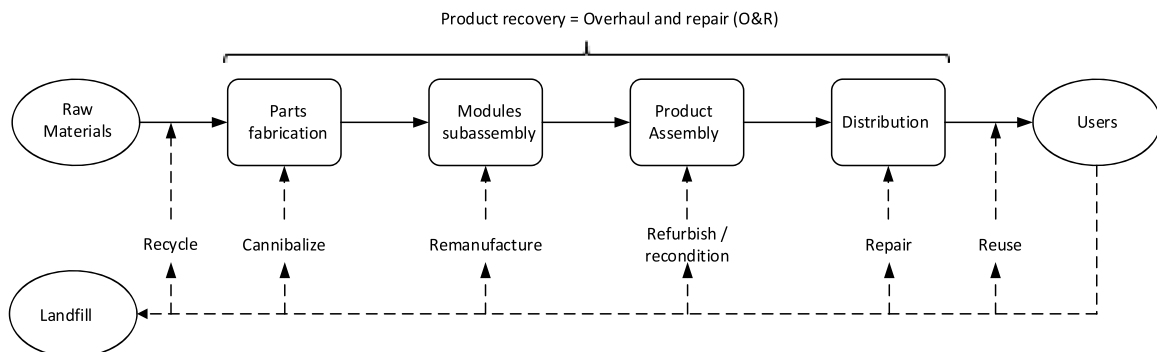


Fig. 1. The product recovery activities considered in this study.

related to RL.

Following the work of Fleischmann et al. [22], several review papers appeared. For example, Ilgin and Gupta [45] updated the review of Gungor and Gupta [30] up until 1997 to cover studies appearing in 1998–2010 focusing on the development of research in environmentally conscious manufacturing and product recovery and classifying the literature by scope (e.g., RL, Recycling/dissassembly, remanufacturing, etc.) and providing pointers for future research directions. Chan et al. [54] reviewed, summarized, and classified the mathematical problems appearing in the literature (1994–2015) on sustainable manufacturing system management to identify recent research focus of those models (e.g., production planning and control, lot sizing, capacity planning, network design, etc.).

Bazan et al. [52] reviewed the mathematics of RL models that stemmed from the seminal work of Schrady [13] and concluded that the focus of the mathematics during the period (1967–2016) was on EOQ-related costs added to product recovery costs. They identified the decision variables used in the surveyed studies as the production and remanufactured (recovered/repared) batch sizes and how many in a cycle. They did not review all the mathematical modeling in the literature, which is the intention of this paper, but rather the milestones (totaling 6 models) representing four research clusters starting with the work of Schrady [13], then jumping to that of Richter [7,8], and then to Teunter [9], followed by that of El Saadany and Jaber [39,42] to give an overview of how modeling evolved. Bazan et al. [52] wanted to emphasize the need to incorporate environmental and societal issues into the EOQ-based RL models and to divert from cost-based modeling to multi-objective where each sub-objective is a scalar function. They advocated that such an approach would give a more balanced consideration between environmental and societal sub-objectives and inventory-related costs that usually overshadow the other two when total system cost is the performance measure.

This paper differs from the work of Bazan et al. [52] as it reviews the mathematical modeling of the EOQ/EPQ-based models from 1967 (starting from Schrady [13]) to December 31, 2022. This review will provide researchers with sort of a road map of how mathematics evolved to the present day. It will save researchers (e.g., established ones, post-doctoral fellows, and PhD students) time by avoiding the need to review the entire RL literature, as it will provide research pointers of interest.

Review search

We have used Google Scholar in our search rather than Scopus. The rationale is that when you search for the journal (source) title “Naval Research Logistics Quarterly” (NRLQ), where the work of Schrady [13] appeared, the oldest article goes back to 1979. When repeating the search without the word “Quarterly” (NRL), the oldest article goes back to 1973, suggesting that the journal was not indexed in Scopus before then. This was very helpful in covering the period where NRL/NRLQ was not indexed. Our search (04/21/2023) used three keywords: “Schrady”, “EOQ” (or “Economic Order Quantity”), and one from “recovery*/repair/refurbish/recondition/remanufacture*”. The number of results (466) was less than the total number of citations that [13] received (555). We have done the same for other researchers (e.g., Richter [7,8], Teunter [9], El Saadany [39,42]) who developed RL

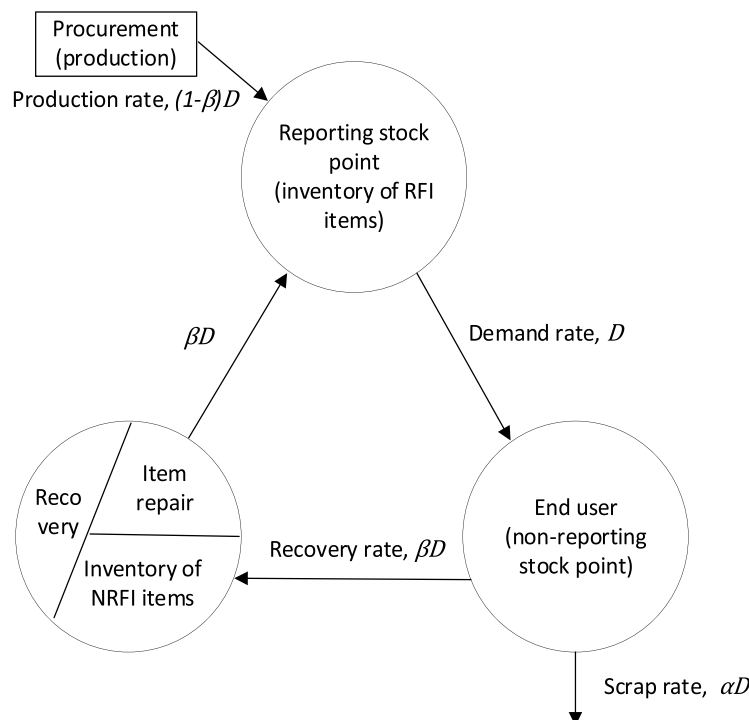


Fig. 2. The repairable inventory system, adapted from [13].

models along the lines of Schrady. With these milestone papers, we then used the “Search within citing articles” feature in Google Scholar for the same set of keywords, i.e., the snowballing method. In addition, we also checked the reference lists of all articles we came across to see if there were older articles that did not cite Schrady [13] but were relevant to our review. We worked through those documents filtering out the following ones:

- non-English
- present/develop non-EOQ-based models
- non-deterministic and fuzzy models
- present/develop models in a supply chain context (joint lot-sizing, multi-echelon)
- not indexed in Scopus
- non-scientific works (e.g., master’s theses under Schrady’s supervision and that extended his work)

Apart from the above, we exclude the models that deal with environmental issues and refer readers to the work of Bazan et al. [52] to see how one could factor them into the mathematics. In other words, the focus will be on the basic structure of the models, i.e., the inventory-related costs, including setup, holding, stockout, disposal, and manufacturing/remanufacturing. The limitation of non-deterministic and multi-echelon models will be handled later in the discussion section.

Our search narrowed down to 32 contributions that are chronologically listed in Table 1. The remainder of this paper is organized as follows. First, the mathematics of EOQ/EPQ models with product recovery are reviewed in the five periods. Second, the limitations of this review are discussed along with the priorities for further research and reviews. Finally, a summary and conclusions are presented.

Mathematics of EOQ/EPQ models with product recovery

The model of Schrady (1967) [13] and those that appeared up until 1990

Schrady’s [13] rationale for his work was that repairable (Schrady used the word *reparable* instead) inventory systems were not adequately studied and that inventory studies focused on the consumable stock. He thought of a repairable system (Fig. 2) to consist of reporting (ready-for-issue; RFI) and non-reporting (end user; non-ready-for-issue, NRFI) stocks and recovery. He described the inventory behavior of a system consisting of one for procurement (production) and the other for repair. Schrady [13] proposed two policies. The first (continuous supplement) policy initiates overhaul and repair (O&R) when NRFI inventory reaches a certain level. The second (substitution) policy suggests a trade-off between stock held in RFI and NRFI by determining the optimum size and number of repair batches in a cycle, supplemented by a procurement batch to compensate for the losses to the system. The continuous supplement policy minimizes the NRFI inventory, and the substitution policy minimizes the RFI inventory. In Fig. 2, D is the demand rate (units/unit-time), and β is the fraction of items recovered (repaired) (% of demand), with $\alpha = 1 - \beta$ being the fraction of items scrapped. The total unit-time cost function of the substitution policy, which is of interest to this article, is:

$$C(Q_p, Q_r) = \frac{K_p D \alpha}{Q_p} + \frac{K_r D \beta}{Q_r} + \frac{h_p \beta}{2} \left[Q_r + \frac{\alpha}{\beta} Q_p \right] + \frac{h_r \beta}{2} (Q_p + Q_r) \quad (1)$$

where Q_p and Q_r are the procurement (production) and repair quantities, K_p and K_r are the fixed setup costs (\$/run), h_p and h_r are the holding costs (\$/unit/unit-time) for RFI and NRFI, respectively. Using differential calculus, Schrady obtained the optimal order quantities of Q_p and Q_r as:

$$Q_p^* = \sqrt{\frac{2K_p D \alpha}{h_p \alpha + h_r \beta}} \quad (2)$$

$$Q_r^* = \sqrt{\frac{2K_r D}{h_p + h_r}} \quad (3)$$

Note that when the scrap rate is zero, i.e., $\alpha = 0$, there is no need to procure (produce) new items as $Q_p = 0$, from Eq. (2). This suggests that Q_p and Q_r should be linked when increasing one would decrease the other, which is not the case in Eq. (1), when, we believe, it should. Schrady numerically showed that the substitution policy (Eq. 1) is more cost-efficient than the continuous supplement policy. The continuous supply of repaired RFIs with new procurement was presented for the first time in a technical paper by Hatchett et al. [58].

The first extension to the work of Schrady [13] is the work of Nahmias and Rivera [27], which considered, like [13], a system of three inventory types. The first two are new procurements and those requiring repair, while the third type is for items beyond repair that must leave the system and are, therefore, not considered a system state. They relaxed the assumption of instantaneous repair rate in [13] by considering a finite rate instead, i.e., $R > 0$, and $R > D$. Their model has m repair batches following a single procurement batch in a cycle, where $m = (\beta/\alpha)Q_p/Q_r$. They showed that when the repair rate is infinite (instantaneous like in Schrady), the unit-time cost function in Eq. (1) could be reduced by $(h_p + h_r)(\beta D/2R)Q_R$. Their closed-form solution differed in Eq. (3), which became:

$$Q_r^* = \sqrt{2K_r D / [(h_p + h_r)(1 - D/R)]} \quad (4)$$

They explored two potential extensions to their model. One extension considers the cost of procurement and repair, i.e., adding the term $c_p D\alpha + c_r D\beta$, with c_p and c_r being the unit procurement and repair costs, respectively. Another extension is to consider floor space constraints for the supply and repair shops, i.e., $a_p Q_p \leq S_p$ and $a_r[\beta(Q_p + Q_r) - Q_r(\beta D/R)] \leq S_r$ with a_p (a_r) is the area needed to store a procured (repaired) item and S_p (S_r) is the total space available for procured (repaired) items.

Models from 1991 to 2000

Mabini et al. [28] noted that earlier work on the topic did not consider the interaction between procurement (production/acquisition) and repair, where perfect recoverability has been assumed, i.e., failed items are repaired and held in the serviceable (RFI) stock. Non-serviceable items are scrapped. They [28] noted that there are situations where items are disposed of, although deemed repairable, resulting in purchasing new items to compensate for lost inventory, allowing procurement and repair to interact. They formulated an EOQ model for a single-item system subject to a stockout constraint. Their unit-time cost is written as:

$$C(Q_p, Q_r, y) = \frac{K_p D\alpha}{Q_p} + \frac{K_r D\beta}{Q_r} + \frac{kc_s}{2} \left[\alpha \frac{(Q_p - y)^2}{Q_p} + \beta \frac{(Q_r - y)^2}{Q_r} \right] + \frac{kc_d}{2} (Q_p + Q_r) \quad (5)$$

where those parameters (now they refer to setup as ordering) not defined above are c_s unit-cost (value) of a serviceable stock, c_d unit-cost (value) of a non-serviceable (recoverable/repairable) stock, k is the cost of capital computed as a percentage of the item unit cost, and y the inventory level signifying stockout. In Eq. (5), the authors in [28] assumed that the unit costs of producing (new), c_p , and that of remanufacturing/repairing (used/returned/collected), c_r , is the same, probably for simplicity, and used the unit cost, c_s , to replace both, i.e., $c_s = c_p = c_r$. If they were to assume that $c_p \neq c_r$, then the third term in Eq. (5) was to be split into two, one term multiplied by c_p and the other by c_r . Mabini et al. [28] minimized Eq. (5) subject to $6y^2(\alpha/Q_p + \beta/Q_r) \leq S$ with S being the maximum allowable unit-time item-periods of stock shortage, $Q_p \geq y$, $Q_r \geq y$, and $Q_p, Q_r, y \geq 0$ and integer. They used a non-linear program optimizer to find the optimal solution. The other works, including Schrady [13] and Nahmias and Rivera [27] and most works that appeared later, did not impose an integer restriction on the production and remanufacturing quantities; they instead assumed real numbers that some rounded to the closest integers. They [28] also provided a version of the model in Eq. (5) for a multi-item system. Interested readers may refer to [28] for details of the mathematics. Note that some papers that fall along the research line of Schrady [13] did not cite his work; instead, those papers cited Nahmias and Rivera [27] and/or Mabini et al. [28]. So, as we move forward in our review, we will search in Scopus for those articles that cited at least one of the three.

With the rise of environmental problems arising from production and consumption, the batch size inventory problem took a twist by focusing on extending the life of products and minimizing waste (scrap) that makes it into landfills. Richter [7,8] described a two-stage EOQ system consisting of two shops, wherein the first items collected from the market during cycle T are repaired to as-good-as-new with the production of new items to fulfill the demand. Collected items are screened and those deemed unrecoverable are considered waste and sent for disposal. He advocated the suitability of using the EOQ model to show the relationship between the waste disposal rate and the optimal production and recovery quantities and, subsequently, the total cost. The total unit-time cost is written as:

$$C(Q, m, n, \alpha) = \frac{(mK_r + nK_p)D}{Q} + \frac{Q}{2} \left[h_s \left(\frac{\alpha^2}{n} + \frac{\beta^2}{m} \right) + h_d \left(\beta + \beta^2 \frac{m-1}{m} \right) \right] \quad (6)$$

where Q is the lot size quantity (new + repair), m and n are the numbers of repair and production (new) setups in a cycle of length T , h_s and h_d are the unit-time holding costs of serviceable and non-serviceable (recoverable/repairable) stocks. Richter [7] minimized Eq. (6) subject to $Q > 0$ and $m, n \in \{1, 2, \dots\}$. The optimal lot size that minimizes Eq. (6) is given as:

$$Q(m, n, \alpha) = \sqrt{\frac{2D(mK_r + nK_p)}{h_s \left(\frac{\alpha^2}{n} + \frac{\beta^2}{m} \right) + h_d \left(\beta + \beta^2 \frac{m-1}{m} \right)}} \quad (7)$$

Richter [7,8] proposed c_p , c_r , and c_w being the unit procurement, repair, and waste disposal costs, respectively, where $\alpha + \beta = 1$, $0 < \alpha, \beta < 1$ of the D . He proposed modifying Eq. (6) to include the following cost term $D[\alpha(c_p + c_w - c_r) + c_r]$. However, Richter [7] noted that the linearity of the suggested cost term could be criticized; also, the absence of setup cost for shifting the used products back to the first shop. He has done some analytical work and provided theorems and proofs. Interested readers can refer to [7,8] for details. Richter [8] showed that small values of the waste disposal rate increase the number of repair setups to a maximum, and large values increase the number indefinitely. Richter [7,8] also found the optimal solution at $m = n = 1$.

In a follow-up paper, Richter [29] continued the analytical study he started in [7,8] by considering extreme waste disposal rates, either $\alpha = 0$ or $\alpha = 1$, to study a bang-bang (pure) policy of either total repair or total waste disposal. The optimality of the pure policies (either repair or disposal of all collected items) holds when the lot size is an integer value. Richter and Dobos [31] extended the discussion in [29] by noting that in the case of linear holding, repair, production, and waste disposal costs with extreme waste disposal rates, the pure strategy to repair or dispose of all used products is optimal. However, they noted that a pure policy is most likely not

technologically feasible as there will always be unrepairable used (collected/returned) items. They advocated that a mixed product recovery policy of production and repair is what practitioners desire. They noted that determining optimal waste disposal rates for changes in waste disposal cost remains a challenging problem.

Teunter et al. [59] cautioned that the traditional way of estimating the holding cost of units in the reverse flow is not as straightforward as in the forward models, i.e., the EOQ/EPQ, which is determined by multiplying the opportunity cost rate by the unit production/purchase cost. They developed and compared different methods for estimating the opportunity cost rates of returned non-serviceable, remanufactured, and manufactured items.

Models from 2001 to 2010

The work of Richter that we discussed in the previous section boosted the research on this topic, with many more papers appearing in 2001–2010 than in 1967–2000.

Teunter [9] studied a deterministic EOQ model where items can be recovered (repaired/refurbished/remanufactured) with the unit holding costs for manufactured (produced) and recovered used items being different while including disposal. The system he considered consisted of manufacturing, a stock of recoverable items, the recovery process, and another stock of serviceable (produced and remanufactured/recovered, see Fig. 1 in [9] on p. 485) items. Used items are collected from customers and moved into the recoverable stock, while the serviceable stock is fed from manufacturing and the recovery process to meet demand. Teunter [9] noted that Richter [7,8] provided a generalized production policy of that of Schrady [13], where he allowed for more than one manufacturing (production) batch in a cycle with disposal occurring at a fixed rate. Teunter [9] found Richter's [7,8] policies sub-optimal since the holding costs are reduced by disposing of returned items in the first segment of the collection interval and collecting items in the second. He generalized Schrady's [13] model by first considering alternating manufacturing, n , and recovery, m , batches, and second by distinguishing the holding cost rates between manufactured and recovered items. Teunter [9] restricted his attention to policies with either $m = 1$ or $n = 1$, as they are optimal (see motivation and proof of Theorem 1, pp. 488–490). Although Teunter assumed that used items are remanufactured to their original "as-new" quality and the unit cost is smaller than the manufacturing cost (i.e., $c_r < c_p$), he mentioned that recovered items have a lower perceived quality than newly produced ones. Unlike Schrady [13], Richter [7,8], and Nahmias and Rivera [27], Teunter [9] assumed that the system could not collect 100% of the sold items, but less than that, i.e., $0 < \gamma = \alpha + \beta < 1$, where γ is the fraction of items (% of demand) returned (collected). The return, recovery, and disposal rates are γD , βD , and αD , respectively. The total cost per unit-time for the policy $m > 0$ and $n = 1$ is given as:

$$C(Q_p, Q_r)_{n=1} = \frac{K_p D(1-\beta)}{Q_p} + \frac{K_r D\beta}{Q_r} + \frac{h_s}{2} \beta Q_r + \frac{h_p}{2} (1-\beta) Q_p + \frac{h_d}{2} \left[\beta Q_r + \left(\beta - \frac{\gamma-\beta}{1-\beta} \cdot \frac{\beta}{\gamma} \right) Q_p \right] + D[(1-\beta)c_p + \beta c_r + (\gamma-\beta)c_w] \quad (8)$$

Note that Teunter assumed three different holding cost rates for manufactured (h_p), recovered (h_s) (both serviceable, but differentiated the two), and returned recoverable (h_d) items, and $h_p > h_s > h_d$. The "returned recoverable" in [9] has the same definition as "non-serviceable" in [7,8], and [28]. The optimal production (manufacturing) and recovery quantities are given, after the first partial derivative of Eq. (8) to zero, as:

$$Q_{p,n=1}^* = \sqrt{\frac{2K_p D(1-\beta)}{h_p(1-\beta) + h_d \left(\beta - \frac{\gamma-\beta}{1-\beta} \cdot \frac{\beta}{\gamma} \right)}} \quad (9)$$

$$Q_{r,n=1}^* = \sqrt{\frac{2K_r D}{h_s + h_d}} \quad (10)$$

with the optimal number or recovery batches as $m^* = (\beta/(1-\beta))Q_{p,n=1}^*/Q_{r,n=1}^*$. Note that in Teunter [9] (p. 491), he has $h_d(1-\beta)$ instead of $h_p(1-\beta)$, which we believe was a typographical error. As for the policy $m = 1$ and $n > 0$, the unit-time cost is the same as Eq. (8) except for replacing the fifth term with $h_d Q_r(\beta/\gamma)/2$, with the optimal production and recovery quantities and the optimal number or production batches are $Q_{p,m=1}^* = \sqrt{2K_p D/h_p}$, $Q_{r,m=1}^* = \sqrt{2K_r D\gamma/(\gamma h_s + h_d)}$, and $n^* = ((1-\beta)/\beta)Q_{r,m=1}^*/Q_{p,m=1}^*$, respectively. Teunter [9] argued that there should never be more than one manufacturing batch nor more than one recovery batch in a cycle, and the optimal ordering quantities for manufacturing and recovery are modifications of the EOQ formula. He found that a mixed policy of disposal and recovery is optimal only if $c_r < c_p^L < c_p < c_p^U$, where c_p^L and c_p^U are some threshold values he found by varying c_p (L = lower, U = upper). Otherwise, dispose (recover) all if $c_p \leq c_p^L$ ($c_p \geq c_p^U$).

Koh et al. [33] focused on the recovery process to find a trade-off between recovery setup costs, ordering cost of new items, and the variable inventory holding costs. In their model, the serviceable and recovery inventories are interdependent, where they determine the optimal recovery and procurement lot sizes for two cases, $R > D$ and $R \leq D$, for two policies: one or more orders for one recovery setup ($m = 1, n \geq 1$), and two or more recovery setups for one order ($m \geq 2, n = 1$). Since Koh et al. [33] have considered part of the system (excluding manufacturing), the mathematics of this model will not be presented here.

In a follow-up paper to [9], Teunter [10] applies the same methodology in [9] to [27]. The total cost per unit-time for the policy $m > 0$ and $n = 1$ is given as:

$$C(Q_p, Q_r) = \frac{K_p D(1-\gamma)}{Q_p} + \frac{K_r D\gamma}{Q_r} + \frac{h_s}{2} [(1-\gamma)(1-D/P)Q_p + \gamma(1-D/R)Q_r] + \frac{h_d\gamma}{2} [Q_p + (1-D/R)Q_r] \quad (11)$$

where $R > D$ is the recovery rate. The optimal production and recovery quantities are determined in an analogous way to Eqs. (9) and (10), and are $Q_{p,n=1}^* = \sqrt{2K_p D(1-\gamma)/[h_s(1-\gamma)(1-D/P) + h_d\gamma]}$ and $Q_{r,n=1}^* = \sqrt{2K_r D/[(h_s + h_d)(1-D/R)]}$, with $m^* = \gamma(Q_{p,n=1}^*/Q_{r,n=1}^*)/(1-\gamma)$. The total cost per unit-time for the policy $m = 1$ and $n > 0$ differs from Eq. (11) in the last (fourth) term, which is replaced with $h_d(1-\gamma D/R)Q_r/2$. Subsequently, $Q_{p,m=1}^* = \sqrt{2K_p D(1-\gamma)/[h_s(1-\gamma)(1-D/P)]}$ and $Q_{r,m=1}^* = \sqrt{2K_r D\gamma/[h_s\gamma(1-D/R) + h_d(1-\gamma D/R)]}$, with $n^* = (1-\gamma)(Q_{r,m=1}^*/Q_{p,m=1}^*)/\gamma$. Teunter [10] reported that his results are valid for finite and infinite production and recovery rates. He also noted that $(m, 1)$ and $(1, n)$ policies could be easily applied in real settings.

Konstantaras and Papachristos [37] extended the two models in Teunter [10] by assuming complete backordering and proposing a rigorous mathematical solution to the problem rather than an approximate one like that of [10]. Their results showed that accounting for backordering makes the system more cost-efficient. They noted that the policies $(m = 1, n > 0)$ and $(m > 0, n = 1)$ are not globally optimal, and more work is needed to show that $(m > 0, n > 0)$ is the globally optimal policy.

Dobos and Richter [35] examined a production/remanufacturing and inventory system (they used the term “recycling”, which is misleading) assuming that remanufactured items are as-good-as-new. They assumed a cycle consists of m remanufacturing (recovery) and n manufacturing (production) batches. In their model, the collection of recoverable items for the next cycle starts after the stock of recoverable items for the current cycle has been depleted and the time required to dispose of unrecoverable items has passed. Their optimal unit-time cost is given as $C(m, n, \gamma, \beta) = \sqrt{2D(mK_r + nK_p)V(m, n, \gamma, \beta)}$, where $V(m, n, \gamma, \beta) = (h_s + h_d)(1-D/R)\gamma^2\beta^2/m + h_s(1-D/P)(1-\gamma\beta)^2/n + h_d\gamma(1-\gamma)\beta^2$, and the optimal cycle time as $T(m, n, \gamma, \beta) = T = \sqrt{2(mK_r + nK_p)/D \cdot V(m, n, \gamma, \beta)}$. The number of units disposed in a cycle is $(1-\gamma)\beta DT$. Dobos and Richter [35] obtained similar results to those in Richter [7] and Dobos and Richter [32], which is a special case of [35] where $m = n = 1$. They asked, “Who must control the quality of the returned items?” and added that not all returned items are recoverable, which they addressed in a follow-up paper [36] where they investigated the model in [35] to two strategies: (1) repurchase all used items D and recover (they used the term “reuse”, which is misleading) maximal ψD (good quality) where $0 < \psi < 1$, or (2) buy back only ψD products and then decide how many of those to recover. Their results suggested that it is better to ‘outsource’ the quality control and repurchase only recoverable products (Strategy 2); a mixed policy of production and recovery (remanufacturing) would be more economical when compared to a bang-bang policy.

Some models address the sequencing of lots within a production cycle, which could influence the inventory holding cost since holding costs of recoverable items and serviceable products are usually unequal, or $h_p > h_r$, and it is more economical to recover than purchase. Choi et al. [38] generalized the Teunter [9] model by treating the sequences of setups for recovered (R) and orders for newly purchased (P) items within a cycle as decision variables; i.e., for example, instead of RRRPP, one might investigate RPRPP, PRRPP, etc. to find that changing the sequence could lower costs. They found that with the maximum deviation of 0.2% from the optimal policy (m, n) , one may as well opt for $(1, n)$ and $(m, 1)$. As this is not a significant finding and contribution since [9], we exclude the mathematics from this review.

El Saadany and Jaber [39] addressed the work of Richter [7,8] and modified the model in two ways. First, they considered the first-time interval that contains no repair (recovery) batches (and not a general interval like that in Richter), where they accounted for the first interval results in residual inventory that overestimates the holding costs. Second, they accounted for switching (changeover) costs when the process alternates between repair and production batches. Switching costs may include, for example, loss of production, additional labor, and alterations to quality. Including the first interval without recovery batches results in the second component of the second cost term in Eq. (6) [7,8]; i.e., $h_d\beta^2(m-1)/m$ is subtracted rather than added. They also proved that the cost $C(Q, m, n, \alpha)$ of a policy whose m and n are even integers can never be optimal. The unit-time cost with switching cost is given as:

$$C(Q, m, n, \alpha) = \frac{[(m-1)K_r + K_{r,1} + (n-1)K_p + K_{p,1}]D}{Q} + \frac{Q}{2} \left[h_s \left(\frac{\alpha^2}{n} + \frac{\beta^2}{m} \right) + h_d\beta \left(1 - \beta \frac{m-1}{m} \right) \right] \quad (12)$$

where $K_{r,1}$ as $K_{p,1}$ are the setup and switching costs of the first repair and the first production runs. Eq. (12) has a unique minimum expressed as:

$$Q(m, n, \alpha) = \sqrt{\frac{2D[(m-1)K_r + K_{r,1} + (n-1)K_p + K_{p,1}]}{h_s \left(\frac{\alpha^2}{n} + \frac{\beta^2}{m} \right) + h_d\beta \left(1 - \beta \frac{m-1}{m} \right)}} \quad (13)$$

They also proved that accounting for switching costs makes an optimal policy with both m and n being even integers valid for some specific conditions.

Drawing an analogy that the behavior of a production system resembles that of physical systems, Jaber and Rosen [40] built a thermodynamic version of Richter’s model [7,8] by applying the first and second laws to improve system performance by reducing system entropy. A follow-up work by Jaber et al. [60] used a similar approach to the EOQ model and showed that ordering larger lots is cheaper than ordering smaller lots. They modeled the demand rate as a heat flow equation, where reducing price (increasing temperature) p below the market (competitors’ price) p_0 creates a product (heat) flow from the system (high-temperature reservoir) to the market (low-temperature reservoir); $D = -\kappa(p - p_0)$ with κ being demand function elasticity similar to thermal capacity in the heat flow equation. This flow generates entropy that needs to be reduced. Similarly, Jaber and Rosen [40] modeled the collection rate of

used items from the market as a commodity flow with the price offered by the system being higher than the market price for a used item, i.e., $v > v_0$ where $D_r = \kappa_r(v - v_0)$ with κ_r being the elasticity of the demand for returned items D_r . For simplicity, they assumed D_r to be a fraction of D , as in the literature. As a result, they added the term $(1 + \beta)\kappa pp_0/Q$ to Eq. (6), with Eq. (7) becoming:

$$Q(m, n, \alpha) = \sqrt{\frac{2[D(mK_r + nK_p) + (1 + \beta)\kappa pp_0]}{h_s\left(\frac{\alpha^2}{n} + \frac{\beta^2}{m}\right) + h_d\left(\beta + \beta^2\frac{m-1}{m}\right)}} \quad (14)$$

Their results showed that a mixed strategy ($m > 0, n > 0$), where $m \neq n$ and noted that entropy cost, could be more suitable for low-demand expensive items than high-demand and inexpensive ones.

The above-surveyed works assumed that demand is satisfied from the serviceable stock containing newly produced and remanufactured items on the premise that both are of equal quality and perceived the same by customers. Jaber and El Saadany [41] addressed this limitation and considered the case of customers perceiving remanufactured items as less valuable, resulting in these items selling for a lower price in a secondary market with its demand rate. This assumption results in lost sales situations for manufactured (newly produced) and remanufactured items over the remanufacturing and production segments of a cycle, respectively. The unit-time cost function for the case of total lost sales is given as:

$$C(m, n, \gamma_p, \gamma_r) = \frac{1}{Q_p/D_p + Q_r/D_r} \left\{ mK_r + nK_p + \frac{h_p}{2nD_p}Q_p^2 + \frac{h_r}{2mD_r}Q_r^2 + c_p\frac{D_p}{D_r}Q_r + c_r\frac{D_r}{D_p}Q_p \right. \\ \left. + h_d \left[\frac{m(\gamma_r\beta_r - 1) + 1}{2mD_r}Q_r^2 + \frac{\gamma_p\beta_p}{2D_p}Q_p^2 + \left(\frac{\gamma_r\beta_r}{mD_p} + \frac{\gamma_p\beta_p(m-1)}{mD_r} \right) Q_pQ_r \right] \right\} \quad (15)$$

where γ_i , D_i , β_i , and c_i are the collection rate, demand rate, recovery rate, and unit cost for item i , where $i = p, r$ (p = production, r = recovery). Note that $c_pD_pT_p + c_rD_rT_r$ is the total lost sales cost in $T = T_p + T_r = Q_p/D_p + Q_r/D_r$, where T_p and T_r are the lengths of production and remanufacturing intervals, respectively. In their second model, Jaber and El Saadany [41] assumed that customers receive compensation to settle for substituting newly produced items for remanufactured ones at some cost. As a result, they modified their demand functions for newly produced and remanufactured items before incorporating them into Eq. (15). The results of their first model showed that the collection of used items from the secondary market, under some conditions, is not optimal. In that case, demand for remanufactured items is satisfied from the primary market. The results of the second model suggested a reduction in total cost when compensation is paid to reduce the effects of lost sales, and the compensation amount holds an optimal value.

El Saadany and Jaber [42] presented the first model for the return rate as a demand function dependent on the offered price and acceptable quality index of a returned item. They extended the work of Dobos and Richter [35], a more general model than Dobos and Richter [34], who considered $m = n = 1$, in this direction and adopted a price-quality demand function from Vörös [61], who modeled demand for a forward flow system. El Saadany and Jaber [42] switched the logic of Vörös's equation to adopt a backward flow, i.e., from the market to the system. The unit-time cost for the case when ($m > 0, n > 0$) is given as:

$$C(m, n, v, q, T) = \frac{mK_r + nK_p}{T} + \frac{TD}{2} G(m, n) + X(v, q) [q(c_r - c_w - c_p - c_m) + c_w + vc_m] + D(c_p + c_m) \quad (16)$$

where $G(m, n) = h_s[\lambda^2(1 - D/R)/m + (1 - \lambda)^2(1 - D/P)/n] + h_d\lambda[1 + \lambda(1 - D/R - m)/m]$, $0 < \lambda = qX(v, q)/D < 1$, $0 < v < 1$ is the (scaled) purchase price of a used returned/collected item, as a percentage of cost of raw materials to produce a new item, $0 < q < 1$ is the quality index, c_m is unit cost of raw materials to produce one unit of a new item, and $X(v, q) = D(1 - Ae^{-\theta v})Be^{-\varphi q}$ is the return (collection) rate, where A , B , θ , and φ are parameters. Their numerical results showed that a mixed policy of production and repair/remanufacturing is optimal. This is contrary to the finding of Dobos and Richter [34,35], who found that a bang-bang policy (repair or scrap all) is optimal.

Konstantaras et al. [43] considered a system for recovered and newly purchased items, where used corrected (returned) items are inspected and sorted. Not all items are remanufacturable to as-good-as-new; some are refurbished and perceived to have lower quality. Those items are sold in a secondary market. Unlike earlier work, this one adopted a profit maximization rather than a cost-minimization approach. Specifically, this paper extends the work of Koh et al. [33] by considering a fixed setup cost and a unit variable cost arising from combining inspection and sorting into one process and by assuming two markets: primary (remanufactured and new) and secondary (refurbished). They considered two policies ($m = 1$ and $n > 0$) and ($m > 0$ and $n = 1$) and investigated two cases. One is when the inspection rate (less those items classified as suitable for refurbishing rather than remanufacturing) is faster than that of demand, which is faster than the return rate, while the other is when it is between demand and the return rate. For the same reason as Koh et al. [33], we will not present its mathematics here. They considered their model to be a generalization of that of [33] in two ways. They viewed inspection and sorting as a single process with a fixed setup cost and a unit variable cost. They also assumed that refurbished items are perceived to be of lower quality than remanufactured or newly purchased ones and, therefore, are sold as a single batch in a secondary market at a reduced price. Their results suggested that a profitable recovery program is one that refurbishes more units than it remanufactures.

Konstantaras and Skouri [44] generalized Teunter [10] model for a policy with a general cycle pattern in which a variable number m of remanufacturing lots of size Q_r is followed by n manufacturing lots each of size Q_p , both are decision variables (cf. sequencing of lots in Choi et al. [38]). They investigated the behavior of the proposed models to minimize the total cost by using numerical examples with parameters collected from the literature.

Models from 2011 to 2020

The work up until 2010 had two main limitations. The first limitation is that a returned used product is recovered as a whole unit, i.e., without considering disassembling the item and extracting value by determining which components to reuse and which items to repair. The second limitation is the assumption of recovering an item infinite times, a technologically infeasible assumption in most cases.

El Saadany and Jaber [46] considered a single product with instantaneous replenishment of manufacturing and remanufacturing inventories, with remanufactured items being as-good-as-new, as assumed in earlier models. Unlike the literature that considers a product as one unit, theirs has a bill-of-material (BOM), and upon returning an item, that item is disassembled, its components screened, and its quality verified. Non-conforming components are disposed of. Those disassembled components are sorted into subassemblies, whose inventory can have three possible behaviors or cases. Each cycle consists of one remanufacturing and one production run. Market demand is replenished from the serviceable stock that could be either new, remanufactured, or hybrid. El Saadany and Jaber [46] were the first to introduce the hybrid policy, building an item using newly purchased, reused, and repaired components. Their cost function is of the form:

$$C(\beta_x, Y_i, V_i) = \sqrt{2D \left[K_r + K_p + \sum_{i=1}^u (Y_i A_i + V_i \hat{K}_{ri}) \right] \times \left[h_r \beta_x^2 + h_p (1 - \beta_x)^2 + \sum_{i=1}^u \hat{h}_i k_i \left[(1 - Y_i) \frac{\beta_x^2}{\vartheta_i} + Y_i \vartheta_i \right] \right]}$$

Subject to:

$$MY_i - (1 - \vartheta_i / \beta_x) \geq 0,$$

$$MV_i - \vartheta_i \geq 0,$$

$$0 < \beta_x < 1,$$

$$Y_i, V_i \in \{0, 1\}, \text{ where } i = 1, 2, \dots, u \quad (17)$$

where D , K_r , K_p , h_r , and h_p are as defined earlier, β_x is ratio of the remanufactured items to total demand and M is a large positive number. For subassembly i , A_i is the ordering cost for a lot, $\hat{K}_{ri}$ is the remanufacturing setup cost, $\hat{h}_i$ is unit per unit-time holding cost, k_i is the number of components, and ϑ_i is the percentage of return. Their results showed that a bang-bang policy (pure production or pure remanufacturing) is optimal for about 65% of these solved examples, with the remaining having a mixed strategy as optimal. This finding suggests that ignoring the BOM and disassembling returned items to salvage and reuse components hides many benefits of product recovery programs.

Jaber and El Saadany [47] investigated the model of Dobos and Richter [34,35] for learning effects in production and remanufacturing (repair), i.e., Shop 1, where new items are produced and collected/returned (used) ones are remanufactured. Shop 2 receives returned items from the market, screens them, disposes of those not repairable, and pushes the repairable ones to Shop 1 for remanufacturing. The buildup of production and remanufacturing inventory is convex since the production and remanufacturing rates become faster due to learning while demand is constant per unit time. Subsequently, the depletion of the repairable stock (Shop 2) follows a concave form. Readers may refer to [62,63] for a review of the effects of learning on lot sizing. Jaber and El Saadany [47] associated learning in production and remanufacturing with investment costs, i.e., $C_{lp} = (1/\Delta_p)\ln(b/b_0)$ and $C_{lr} = (1/\Delta_r)\ln(a/a_0)$, where $\Delta_p(\Delta_r)$ is the percentage increase in the learning exponent, $b(a)$, per dollar invested, for production (remanufacturing) process, with $b_0(a_0)$ being the initial value of the learning exponent. They derived the holding cost in Shop 1 as:

$$H_1(n, m, Q, \beta) = h_p n \frac{Q_p^2}{2D} - h_p \frac{y_1 (nQ_p)^{2-b}}{(1-b)(2-b)} + h_p \frac{y_1 Q_p^{2-b}}{(1-b)(2-b)} \left[(n - 0.5)^{2-b} - 0.5^{2-b} \right] + h_r m \frac{Q_r^2}{2D} - h_r \frac{\omega_1 m^{2-a} Q_r^{2-a}}{(1-a)(2-a)} \quad (18)$$

where y_1 and ω_1 are the initial times to produce and remanufacture the first unit. The other parameters have been defined earlier. They derived the holding cost in Shop 2 as:

$$H_2(n, m, Q, \beta) = h_d \left\{ m \omega_1 Q_r^{2-a} \left[\frac{m^{2-a} - (m-1)^{2-a}}{(1-a)(2-a)} - \frac{(m-1)^{1-a}}{1-a} \right] + \frac{m(m-1)Q_r^2}{2D} + \frac{\beta}{2D} (mQ_r + nQ_p)^2 - \frac{n\beta Q_p}{1-\beta} \left[(m-1) \frac{Q_r}{D} + \frac{\omega_1 Q_r^{1-a}}{1-a} (m^{1-a} - (m-1)^{1-a}) \right] \right\} \quad (19)$$

They derived the production and remanufacturing of nQ_p and mQ_r quantities in a cycle of length T , where $T = (nQ_p + mQ_r)/D$. The total labor cost in T was given as:

$$L(n, m, Q, \beta) = c_p \frac{y_1}{1-b} (nQ_p)^{1-b} + c_r \frac{\omega_1}{1-a} (mQ_r)^{1-a} \quad (20)$$

Note that Jaber and El Saadany [47], for simplicity, assumed no transfer of learning between cycles, i.e., what happens in T duplicates itself indefinitely, whether those cycles are production or remanufacturing. Again, for simplicity, they assumed no knowledge transfer (total forgetting) between remanufacturing and production runs due to product similarities. These two limitations deserve an extension of this work. They then minimized the unit-time cost as:

$$C(n, m, Q, \beta) = \frac{(1-\beta)}{nQ} \{mK_r + nK_p + H_1(n, m, Q, \beta) + H_2(n, m, Q, \beta) + L(n, m, Q, \beta) + C_{lp} + C_{lr}\} \quad (21)$$

Subject to:

$$0 \leq \beta \leq \beta_{max}$$

$$n, m \in \{1, 2, \dots\}$$

$$Q_p > 0$$

where β_{max} is the maximum percentage of the demand rate that is collected for recovery. Their numerical results showed there is a ceiling to how much can be invested to accelerate learning and reap savings, even when y_1 and ω_1 are significantly high. They also showed that faster learning in production lowers the collection rate of used items. Shortening the cycle time will result in fewer collected (used) items being recovered, making a product recovery program unsustainable due to the economics of scale. This observation suggests that optimal production and remanufacturing learning rates exist.

Jaber et al. [48], along the same lines as Jaber and Rosen [40] but with a different approach, accounted for exergy cost in addition to entropy cost and presented a simple RL system. Exergy and entropy are directly related as both are founded on the second law of thermodynamics. Exergy is consumed or destroyed rather than conserved in a process because of its irreversibility and is often quantified as entropy creation. Entropy in a system represents waste, for example, of energy or raw material, and in rework, scrap, and idleness, to name a few, i.e., when using work to produce useful energy or exergy. They introduced a simpler version of an earlier model (Jaber and Rosen [40]) and a new cost component, exergy cost. Because of this similarity, the mathematics of this work will not be presented here.

Hasanov et al. [49] revised Jaber and El Saadany [41], i.e., Eq. (15), who, unlike Richter [7], assumed remanufactured items are sold in a secondary market at a discounted price with unmet demand lost at a cost, and instead of lost sales allowed for backordering, full or partial. Like the literature, they considered a cycle to consist of production runs (for new items) and remanufacturing runs (for recovering collected/used items). Demand for remanufactured items is backordered over the production runs, and that for newly produced items is backordered over the remanufacturing runs. They modified the model in [41] by replacing the total lost sales cost in T with the total backordering cost (BC). The term is of the form $BC_T(m, \gamma_r, \gamma_p) = m^2[s_p\theta_p D_p/D_r^2 + s_r\theta_r(D_r/D_p^2)((1-\gamma_r\beta_r)/\gamma_p\beta_p)^2]/2$, where all the parameters are defined above except for s_p and s_r , the backorder cost for produced and remanufactured items, respectively, and θ_p and θ_r , the proportions of production and remanufacture demand that is backordered, respectively. Total backordering case occurs when $\theta_p = \theta_r = 1$ and partial backordering when $0 < \theta_p, \theta_r < 1$. The total lost sales cost in [41] was modified as $c_p D_p T_p (1 - \theta_p) + c_r D_r T_r (1 - \theta_r)$. They found that the proposed inventory policy can occur when there are several production and remanufacturing batches in an interval and when the sum of the setup costs (including additional components) is significant. Singh et al. [64] extended the work of Hasanov et al. [49] by assuming finite production and remanufacturing rates and considering also the cost of disposing of items. They illustrated the behavior of their model with some numerical examples.

El Saadany et al. [50] gathered evidence from the literature to address a fundamental limitation in earlier works, i.e., the assumption that an item can be remanufactured indefinitely. They have modeled the recoverable percentage of returns, i.e., β , as a function of the number of items to be remanufactured, ζ , where increasing ζ is associated with a concave form investment in design function. Before studying this for the works of Richter [29] and Teunter [9], they replaced β with $\beta_\zeta = 1 - (1 - \beta)(1 - \beta^\zeta + 1)^{-1}$. Their results suggested that assuming unlimited remanufacturing will not capture the intended benefits of product recovery programs. Their results showed that there is an optimal number of remanufacturing generations that balances costs. Those results also showed that as more units are collected ($\beta \uparrow$), continuing with the assumption of infinite remanufacturing worsens the system's performance. Their results implied that extending a product's life through investment in design reduces disposal and uses less material and energy.

Despite some work considering different types of product returns (e.g. [41,43]), Sun et al. [53] addressed a limitation in the previous models, i.e., the assumption that the recovery rates are equal for each item. The items are sorted upon return. Those with the lowest quality are scrapped for materials or parts recovery, and those with higher quality are labelled for remanufacturing. Sun et al. [53] further classified returns for remanufacturing into multiple quality grades where the remanufacturing (recovery) rate increases as the quality rate increases. They assumed that the proportion of each quality grade held a constant value. The optimal manufacturing lot size for what they called a hybrid (production and remanufacturing) system was expressed as $Q^* = \sqrt{\frac{2(A_R + A_M)}{(h_p - h_r)i_s(D_r, R, P, q)}}$, where A_R and A_M are ordering costs of returns and raw materials, and $i_s(D_r, R, P, q)$ is a function of demand rate of remanufactured products, recovery rate, manufacturing rate, and quality in the returns for remanufacturing. Function i_s depends on the remanufacturing sequence. The remanufacturing line can only handle one grade at a time, so scheduling is important. One may take advantage of the low holding cost for low-quality (inexpensive) items and postpone remanufacturing them until they are needed. In addition, in the hybrid

cost for components between production lots, they suggested not buying components in fixed lot sizes. Readers may refer to [55] for mathematical formulation.

Recent work (2021–2022)

Hsieh et al. [56] studied an EPQ model involving the sequential use of raw and recycled materials. Recycled materials are less expensive. However, production setup time with recycled materials becomes uncertain since additional procedures (fine-tuning of the material mix, tooling adjustment, calibration) are implemented. The study assumed an upper limit for setup time for recycled materials to alleviate prolonged risk. Hsieh et al. [56] numerically showed that the upper setup time value and lot size of the recycled materials can serve as cost reduction drivers for the manufacturer of a prolonged setup time.

Piñeyro and Viera [57] addressed an economic lot-sizing problem with remanufacturing assuming a finite set of quality levels for grading returns, each related to a remanufacturing type and task. Better quality increases inventory costs (more expensive items) but reduces remanufacturing costs. Their numerical results showed that all the available returns of the same quality should be remanufactured in some periods, but not necessarily those of the highest quality.

Discussion and concluding remark

In this section, we first check the accuracy of our literature review using the connected papers tool (Section 3.1). Then, Section 3.2 briefly discusses the limitations, i.e., the modeling approaches that were out of the scope of this review, followed by a more in-depth discussion of environmental and societal factors in Section 3.3. Section 3.4 suggests other research directions, followed by a concluding remark in Section 3.5.

Connected papers check

Our review started with the seminal work of Schrady [13], the first deterministic EOQ model with product recovery, published in a scientific journal. Following this line of research, we used the milestone papers of Richter [7,8], Teunter [9], and El Saadany [39,42] to explore how the mathematics of the models evolved. Afterwards (06/05/2023), we checked the accuracy of our review by placing Schrady [13] as the "origin" paper in the connected papers [65], a free tool to explore research papers. The graph in Fig. 3 arranges papers by similarity, i.e., based on co-citation and bibliographic coupling, i.e., two papers having highly overlapping citations and references are presumed to have a higher chance of treating a related subject matter. Fig. 3 confirms that our review (results shown in Table 1) considered all the papers closely related to Schrady's work. Not all the papers presented in Fig. 3 contribute to or follow the present line of research in modeling.

Limitations

This review focused on deterministic inventory models. In real life, market behavior, item failure, delivery lead times, and various disruptions are, at least to some extent, unpredictable. Therefore, stochastic models have been developed alongside deterministic ones. Reviewing the extensive number of stochastic mathematical EOQ models for RL deserves a stand-alone paper. For this, interested researchers could start from the two review papers, the one covering the works conducted under Schrady's supervision in the Naval Postgraduate School [25] and the other that covers those models that appeared by 1991 [66].

In recent decades, lot-sizing with RL has expanded into multi-echelon models, which were excluded from our review. For example, Hasanov et al. [67] developed a 4-level supply chain by introducing a reverse channel with subassemblies [45]. Their results showed that increasing the collection rate reduces total supply chain costs and the consumption of virgin materials. Those results also recommended reducing storage costs for returned subassemblies and remanufactured items and simplifying a product's bill-of-material for ease of disassembly and recovery to make product recovery programs more cost-effective. Hasanov et al. [67] also showed that increasing the number of times a returned used item is remanufactured does not substantially increase emissions and energy costs. Bazan et al. [68], Tahirov et al. [69], and Hasanov et al. [67] all found that increasing the collection rate could make a CLSC more cost-effective, competitive, and sustainable.

Environmental and societal factors

RL models taking on environmental factors started to appear in the literature following the work of Bonney and Jaber [1], which was the first to discuss the need for designing environmental production and responsible inventory systems. Next, we discuss some of the models in this line of research.

Hasanov et al. [70] extended the work of Richter [7] to account for energy, transportation, and waste disposal costs to factor in environmental sustainability by using a life cycle cost (LCC) approach. This work is probably the first to introduce environmental issues into the RL framework put forward by Schrady [13] and later Richter [7]. The energy cost parameters used in the LCC approach consider social costs (human health, pollution, etc.) to analyze different energy generation systems (renewable or non-renewable) that are not reflected in the price of electricity. They assumed the social costs to vary by the type and amount of pollutants emitted by the energy technology used to generate electricity. LLC is the sum of capital cost (initial capital expense for equipment, system design, system engineering and installation), operation and maintenance costs (salaries, inspection, insurance, etc.), fuel costs, and external

costs (e.g., damage prevention or damage costs) less the salvage value of the system. The results showed that accounting for energy, transportation, and waste disposal costs significantly affects the production, remanufacturing, and inventory policies. They also showed that using a renewable or a mixed energy source has economic and social benefits much more than using only a non-renewable energy source.

Bazan et al. built on the work of Richter [7] and El Saadany et al. [50] by allowing for multiple manufacturing and remanufacturing cycles and adding the costs of GHG emissions generated from manufacturing, remanufacturing, and transportation of items, and that from energy used in producing new and remanufactured items. Their proposed model is a first step towards developing an environmentally responsible RL inventory model. Their results showed that optimizing the manufacturing/remanufacturing and inventory-related costs and emissions and energy costs promotes less remanufacturing, as opposed to just focusing on solid waste disposal, which is what earlier studies have focused on. They also showed the benefits of collecting more units for remanufacturing. We refer the reader to Bazan et al. [52] for insights on the benefits of incorporating environmental issues into RL and how to modify mathematical models to capture them.

Hasanov et al. [70] and Bazan et al. [68] are the earliest works that brought environmental issues into an RL context, i.e., building along the lines of Schrady [13] and Richter [7,8]. Bazan et al. [52] focused on reviewing four mathematical models (representing clusters, i.e., Schrady [13], Richter [7,8], Teunter [9], and El Saadany [39,42]) on RL models that stemmed from the work and noted that the mathematics after 60+ years of modeling development continues to use EOQ/EPQ related costs and some recovery process costs. Despite the rise in environmental awareness, they stressed that the modeling in the identified clusters ignored pressing issues like reducing GHG emissions and energy consumption from RL operations and suggested a possible approach to modeling environmentally responsible EOQ/EPQ-based RL models. Their suggested approach minimizes multiple objectives, scaling each sub-objective to a reference value, including classic EOQ/EPQ RL-related costs and individual environmental factors. They advocated that this is more appropriate than using costs since the latter for some environmental factors are difficult to estimate. Their objective function took the form $F = w_1 \left(\frac{RL \text{ System Cost}}{Reference \text{ Cost}} \right) + w_2 \left(\frac{GHG \text{ Emissions}}{Reference \text{ Emissions}} \right) + w_3 \left(\frac{Energy \text{ Consumed}}{Reference \text{ Energy}} \right) + w_4 \left(\frac{Solid \text{ Waste}}{Reference \text{ Waste}} \right)$, where $\sum_{i=1}^4 w_i = 1$. In their suggested model, for example, if management focuses only on reducing EOQ/EPQ RL-related costs, GHG emissions, energy consumption, and waste disposal will increase. If the focus is on minimizing GHG emissions, EOQ/EPQ RL-related costs would increase, and energy consumption could either increase or decrease, with no change to solid waste disposal quantities. They cautioned that their suggested modeling approach is ‘only’ for illustrative purposes without attempting to solve it. They implicitly encouraged interested researchers to investigate this modeling approach further in future work.

The work of Bazan et al. [52] triggered a research stream that accounts for environmental issues in RL and CLSC systems, as reflected in the number of citations it has received. A few of the recent articles that have appeared following the work of Bazan et al. [52] are briefly reviewed here for the interest of the readers. For example, Chen and Bidanda [71] modeled an EPQ-based CLSC for a beer manufacturer that collects empty bottles that are checked, cleaned, disinfected, and filled before sending them back to the market. Bottles identified as non-reuseable are collected and recycled by a third party, with the missing quantity of needed bottles replenished by a supplier. They showed that a cap-and-trade policy was preferred by all system partners as it reduces carbon emissions without a noticeable increase in system cost. Liao et al. [72] investigated an EOQ-based CLSC system from two perspectives, resulting in two models. The first EOQ model focuses on environmental sustainability (ES-EOQ) that reduces carbon emissions from operational activities without additional investment. The second EOQ model balances costs and environmental sustainability (BAS-EOQ). They found that fluctuations in carbon trading prices affect the environmental efficacy of the BAS-EOQ policy/model. They also found that an ES-EOQ model/policy would result in reducing carbon emissions for an increase in system cost. These findings and those of Chen and Bidanda [71] corroborate the need for the modeling approach suggested by Bazan et al. [52]. Condeixa et al. [73] proposed a sustainable EOQ model for RL for remanufacturing inventory and disposal decisions. They found that carbon prices significantly impact remanufacturing taxes and, socially, poor working conditions can lower product recovery rates and impact the sustainability of RL systems.

Other research directions

The literature has also shown some studies that took the RL research into new directions. Some of those studies will be reviewed next.

Having enough returns to remanufacture a new product may not always be possible. Seeding, where a firm sells a new product as remanufactured at the start of its lifecycle to increase core recovery quantities, could be a solution to satisfy the demand for remanufactured [74]. The rationale is to keep reaching customers unwilling to pay for pricy new items and meet the increasing demand for remanufactured items. To encourage returns of its own or its competitors’ products, the acquisition price a CLSC offers must entice buyers and ensure the quality of returns, and the extracted value of disassembled items must be acceptable [75,76]. Despite the realized benefits of seeding, it has not been investigated in a CLSC context. Managers lack tangible tools to assess the benefits of seeding and manage the process. Apart from wear and tear, the quality of used items depends on the quality of the original items, which suggests a promising new research direction focusing on the quality of both the forward (product) and reverse (core).

The reviewed mathematical models assume that returns represent a fraction of constant product demand and that raw materials, components, and returned items are delivered to the manufacturer on time with the exact quantities and quality. This assumption is severely limited as it diverges from reality [77]. Cores to be remanufactured must also meet certain quality levels and competitive pricing but are often returned at different random times with varying quantities. Finding strategies to increase core collection to enhance remanufacturing profitability is not enough without considering the delivery performance model [77] to effectively

synchronize the forward flow of materials and items and reverse flow of cores for remanufacturing to facilitate optimal production planning and control, which is also a promising research direction to pursue.

The earliest attempt to account for the randomness of core availability in EOQ/EPQ-based RL models was by Schradly [25]. These models use, e.g., policies to project or “look ahead” at the available core quantity until a recovery reorder point [16]. As in many manufacturing domains, periodic status updates no longer suffice, and the means of capturing real-time data has come to the forefront. Further research should, therefore, explore how data from smart connected products [78] could be utilized in predicting product failure/end-of-life and core returns and how it would affect product recovery programs.

Concluding remark

The concept of reverse logistics (RL) got a boost in the 1990s with the work of Richter and later Teunter due to the rise in societal, environmental, and governmental concerns. The RL concept is based on circular economy principles, which aim to reduce waste and conserve resources by optimizing material and energy flows in closed-loop supply chains. These supply chains play a crucial role in our daily lives, and it is important to develop sustainable production and inventory management systems that prioritize environmental responsibility. This involves designing efficient product recovery and logistics processes that minimize waste and promote resource conservation.

Here, we summarize what we have observed from reviewing the RL mathematical models. Before El Saadany and Jaber [42], 2010, the RL literature ignored the quality and pricing of returned items. They modeled the return (collection) rate of used items as a dependent function of the quality and unit acquisition price. Their results showed that a mixed strategy of producing and remanufacturing is profitable, which is contrary to the literature (e.g., [35]) that favors a bang-bang policy (either manufacturing new or remanufacturing used items, but not both). The literature also assumed that returned used items are remanufactured as whole units, i. e., ignoring disassembly. El Saadany and Jaber [46], in 2011, addressed this key limitation by assuming that collected items are disassembled and their components inspected before they are sorted into subassemblies; each has its return rate and inventory policy. Their approach uncovered the hidden benefits of product recovery programs, where they were the first to introduce the term ‘hybrid’ product composed of reused, remanufactured, and new components. El Saadany et al. [50], in 2013, addressed another key research gap by assuming a finite, rather than an indefinite, number of product recovery times and linking its increase to investment in design for remanufacturing. El Saadany et al. [50] considered neither the quality and price of a returned item [11] nor the subassemblies of components. They showed that collecting more items for recovery is desirable and ignoring the finite remanufacturing case worsens system performance and produces unrealistic results. Tahirov et al. [69], in 2016, framed the model in [46] into a CLSC (manufacturer-retailer-remanufacturer), where a used product and its spare parts are returned for recovery through remanufacturing and recycling. They showed that the collection rate and disassembly of products and spare parts significantly impacted system performance and recommended a mixed strategy (production & remanufacturing) to reduce the heavy reliance on external suppliers and extend the product’s useful life to lower environmental impact. Sun et al. [53], in 2018, addressed another research gap by assuming the recovery rates are non-equal for each item but increase as the quality rate (grade of the cores) increases. They found that, in general, faster recovery rates make scheduling the remanufacturing line beneficial when one type of grade can be handled at a time. The low holding cost for returned low-quality items makes it reasonable to postpone their remanufacturing until they are needed.

In summary, this paper provides a condensed review of the works that stemmed from the work of Schradly [13] and highlights some promising research directions that one of the authors will initiate soon. Although we have tried to be as comprehensive as possible, we have also tried hard to be concise to give readers interested in this research a good enough starting background and save them time avoiding the need to go through the entire literature. Starting with this article, researchers can then choose to delve into one of the surveyed and discussed works. We believe this review will be well-received by researchers in the field and provide valuable research ideas for graduate students and postdoctoral fellows in industrial engineering and related fields (e.g., production and operations management). It was not possible to develop a taxonomy to group the EOQ/EPQ-based RL works and models based on their coverage, assumptions, etc. First, it will extend the paper significantly, which is not desired, and second, it will be out of the intended scope of this paper. However, this could be left for future research. Finally, it was not possible to provide an in-depth critique without industrial data on the collection, inspection, waste, recovery (remanufacturing/repair), quality, costs, etc., that could be used to test some potential models in the literature to see how some of the assumptions made could bring such models close to what one expects to observe in practice and to critique the assumptions that do not hold and deserve further investigation and a standalone paper.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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References

- [1] M. Bonney, M.Y. Jaber, Environmentally responsible inventory models: non-classical models for a non-classical era, *Int. J. Prod. Econ.* 133 (2011) 43–53, <https://doi.org/10.1016/J.IJPE.2009.10.033>.
- [2] M. Klausner, C.T. Hendrickson, Reverse-logistics strategy for product take-back, *Interfaces (Providence)* 30 (2000) 156–165, <https://doi.org/10.1287/INTE.30.3.156.11657>.
- [3] E. MacArthur, Towards the circular economy, *J. Ind. Ecol.* 2 (2013) 23–44, https://www.werktrends.nl/app/uploads/2015/06/Rapport_McKinsey-Towards_A_Circular_Economy.pdf (accessed June 1, 2023).
- [4] M. Geissdoerfer, P. Savaget, N.M.P. Bocken, E.J. Hultink, The circular economy – A new sustainability paradigm? *J. Clean. Prod.* 143 (2017) 757–768, <https://doi.org/10.1016/J.JCLEPRO.2016.12.048>.
- [5] N. Turken, V. Cannataro, A. Geda, A. Dixit, Nature inspired supply chain solutions: definitions, analogies, and future research directions, *Int. J. Prod. Econ.* 58 (2020) 4689–4715, <https://doi.org/10.1080/00207543.2020.1778206>.
- [6] M.C. Bonney, Trends in inventory management, *Int. J. Prod. Econ.* 35 (1994) 107–114, [https://doi.org/10.1016/0925-5273\(94\)90070-1](https://doi.org/10.1016/0925-5273(94)90070-1).
- [7] K. Richter, The EOQ repair and waste disposal model with variable setup numbers, *Eur. J. Oper. Res.* 95 (1996) 313–324, [https://doi.org/10.1016/0377-2217\(95\)00276-6](https://doi.org/10.1016/0377-2217(95)00276-6).
- [8] K. Richter, The extended EOQ repair and waste disposal model, *Int. J. Prod. Econ.* 45 (1996) 443–447, [https://doi.org/10.1016/0925-5273\(95\)00143-3](https://doi.org/10.1016/0925-5273(95)00143-3).
- [9] R.H. Teunter, Economic ordering quantities for recoverable item inventory systems, *Nav. Res. Logist.* 48 (2001) 484–495, <https://doi.org/10.1002/nav.1030>.
- [10] R. Teunter, Lot-sizing for inventory systems with product recovery, *Comput. Ind. Eng.* 46 (2004) 431–441, <https://doi.org/10.1016/j.cie.2004.01.006>.
- [11] A. Andriolo, D. Battini, R.W. Grubbström, A. Persona, F. Sgarbossa, A century of evolution from Harris's basic lot size model: survey and research agenda, *Int. J. Prod. Econ.* 155 (2014) 16–38, <https://doi.org/10.1016/J.IJPE.2014.01.013>.
- [12] M.A. Bushuev, A. Guiffrida, M.Y. Jaber, M. Khan, A review of inventory lot sizing review papers, *Manag. Res. Rev.* 38 (2015) 283–298, <https://doi.org/10.1108/MRR-09-2013-0204/FULL/PDF>.
- [13] D.A. Schrady, A deterministic inventory model for repairable items, *Nav. Res. Logist. Q.* 14 (1967) 391–398, <https://doi.org/10.1002/NAV.3800140310>.
- [14] E. Silver, D. Pyke, R. Peterson, Inventory management and production planning and scheduling, 1998, https://www.academia.edu/download/49393751/Inventory_Management_and_Production_Plan20161006-20229-123n7yx.pdf (accessed June 1, 2023).
- [15] A.M. King, S.C. Burgess, W. Ijomah, C.A. McMahon, A.M. King, Reducing waste: repair, recondition, remanufacture or recycle? *Wiley Online Libr* 14 (2006) 257–267, <https://doi.org/10.1002/sd.271>.
- [16] M. Hoekstra, Supply management models for repairable items, *Fifth U.S. Army Oper. Res. Symp.* (1966) 375–395, <https://apps.dtic.mil/sti/tr/pdf/AD0815026.pdf> (accessed February 4, 2022).
- [17] S. Nahmias, Managing repairable item inventory systems: a review, *TIMS Stud. Manag. Sci.* 16 (1981) 253. –177.
- [18] S. Axsäter, Continuous review policies for multi-level inventory systems with stochastic demand, *Handb. Oper. Res. Manag. Sci.* 4 (1993) 175–197, [https://doi.org/10.1016/S0927-0507\(05\)80184-7](https://doi.org/10.1016/S0927-0507(05)80184-7).
- [19] E. van der Laan, The effects of remanufacturing on inventory control, 1997, <https://repub.eur.nl/pub/14828/CHAPTER4.PDF> (accessed June 1, 2023).
- [20] D.P. Heyman, Optimal disposal policies for a single-item inventory system with returns, *Nav. Res. Logist.* 24 (1977) 385–405, <https://doi.org/10.1002/NAV.3800240302>.
- [21] J.A. Muckstadt, M.H. Isaac, An analysis of single item inventory systems with returns, *Nav. Res. Logist. Q.* 28 (1981) 237–254, <https://doi.org/10.1002/NAV.3800280207>.
- [22] M. Fleischmann, J.M. Bloemhof-Ruwaard, R. Dekker, E. Van Der Laan, J.A.E.E. Van Nunen, L.N. Van Wassenhove, Quantitative models for reverse logistics: a review, *Eur. J. Oper. Res.* 103 (1997) 1–17, [https://doi.org/10.1016/S0377-2217\(97\)00230-0](https://doi.org/10.1016/S0377-2217(97)00230-0).
- [23] V.D.R. Guide, R. Srivastava, Repairable inventory theory: models and applications, *Eur. J. Oper. Res.* 102 (1997) 1–20, [https://doi.org/10.1016/S0377-2217\(97\)00155-0](https://doi.org/10.1016/S0377-2217(97)00155-0).
- [24] M.A. Bushuev, A. Guiffrida, M.Y. Jaber, M. Khan, A review of inventory lot sizing review papers, *Manag. Res. Rev.* 38 (2015) 283–298, <https://doi.org/10.1108/MRR-09-2013-0204>.
- [25] D.A. Schrady, Mathematical models of the repairable item inventory system, Monterey, California, 1967, <https://core.ac.uk/download/pdf/36718913.pdf> (accessed March 24, 2022).
- [26] M. Mabini, L. Gelders, Repairable item inventory systems: a literature review, *JORBEL Belgian J. Oper. Res. Stat. Comput. Sci.* 30 (1990) 57–69, <https://www.orbel.be/jorbel/index.php/jorbel/article/view/106> (accessed February 9, 2022).
- [27] S. Nahmias, H. Rivera, A deterministic model for a repairable item inventory system with a finite repair rate, *Int. J. Prod. Res.* 17 (1979) 215–221, <https://doi.org/10.1080/00207547908919609>.
- [28] M.C. Mabini, L.M. Pintelon, L.F. Gelders, EOQ type formulations for controlling repairable inventories, *Int. J. Prod. Econ.* 28 (1992) 21–33, [https://doi.org/10.1016/0925-5273\(92\)90110-S](https://doi.org/10.1016/0925-5273(92)90110-S).
- [29] K. Richter, Pure and mixed strategies for the EOQ repair and waste disposal problem, *OR Spectrum* 19 (1997) 123–129, <https://doi.org/10.1007/BF01545511>.
- [30] A. Gungor, S.M. Gupta, Issues in environmentally conscious manufacturing and product recovery: a survey, *Comput. Ind. Eng.* 36 (1999) 811–853, [https://doi.org/10.1016/S0360-8352\(99\)00167-9](https://doi.org/10.1016/S0360-8352(99)00167-9).
- [31] K. Richter, I. Dobos, Analysis of the EOQ repair and waste disposal problem with integer setup numbers, *Int. J. Prod. Econ.* 59 (1999) 463–467, [https://doi.org/10.1016/S0925-5273\(98\)00110-8](https://doi.org/10.1016/S0925-5273(98)00110-8).
- [32] I. Dobos, K. Richter, The integer EOQ repair and waste disposal model - further analysis, *Cent. Eur. J. Oper. Res.* 8 (2000) 173–194, <http://search.ebscohost.com/login.aspx?direct=true&profile=ehost&scope=site&authtype=crawler&jrnl=1435246X&asa=Y&AN=9169921&h=8nbxh8T8J7AbOm5q0Z6tDMUSnVk%2BBPTUsM%2Bt8I8RzmfFEEqYUkDmu7gLgVXKW9a3Qwe63AXe2XD8VuBqrvA%3D%3D&crl=c> (accessed February 9, 2022).
- [33] S.G. Koh, H. Hwang, K.I. Sohn, C.S. Ko, An optimal ordering and recovery policy for reusable items, *Comput. Ind. Eng.* 43 (2002) 59–73, [https://doi.org/10.1016/S0360-8352\(02\)00062-1](https://doi.org/10.1016/S0360-8352(02)00062-1).
- [34] I. Dobos, K. Richter, A production/recycling model with stationary demand and return rates, *Cent. Eur. J. Oper. Res.* 11 (2003) 35–46, <https://core.ac.uk/reader/147610080> (accessed February 9, 2022).
- [35] I. Dobos, K. Richter, An extended production/recycling model with stationary demand and return rates, *Int. J. Prod. Econ.* 90 (2004) 311–323, <https://doi.org/10.1016/j.ijpe.2003.09.007>.
- [36] I. Dobos, K. Richter, A production/recycling model with quality consideration, *Int. J. Prod. Econ.* 104 (2006) 571–579, <https://doi.org/10.1016/J.IJPE.2005.09.006>.
- [37] I. Konstantaras, S. Papachristos, Lot-sizing for a single-product recovery system with backordering, *Int. J. Prod. Res.* 44 (2006) 2031–2045, <https://doi.org/10.1080/00207540500380924>.
- [38] D.W. Choi, H. Hwang, S.G. Koh, A generalized ordering and recovery policy for reusable items, *Eur. J. Oper. Res.* 182 (2007) 764–774, <https://doi.org/10.1016/j.ejor.2006.08.048>.
- [39] A.M.A. El Saadany, M.Y. Jaber, The EOQ repair and waste disposal model with switching costs, *Comput. Ind. Eng.* 55 (2008) 219–233, <https://doi.org/10.1016/j.cie.2008.01.016>.

- [40] M.Y. Jaber, M.A. Rosen, The economic order quantity repair and waste disposal model with entropy cost, *Eur. J. Oper. Res.* 188 (2008) 109–120, <https://doi.org/10.1016/j.ejor.2007.03.016>.
- [41] M.Y. Jaber, A.M.A. El Saadany, The production, remanufacture and waste disposal model with lost sales, *Int. J. Prod. Econ.* 120 (2009) 115–124, <https://doi.org/10.1016/j.ijpe.2008.07.016>.
- [42] A.M.A. El Saadany, M.Y. Jaber, A production/remanufacturing inventory model with price and quality dependant return rate, *Comput. Ind. Eng.* 58 (2010) 352–362, <https://doi.org/10.1016/j.cie.2009.01.017>.
- [43] I. Konstantaras, K. Skouri, M.Y. Jaber, Lot sizing for a recoverable product with inspection and sorting, *Comput. Ind. Eng.* 58 (2010) 452–462, <https://doi.org/10.1016/J.CIE.2009.11.004>.
- [44] I. Konstantaras, K. Skouri, Lot sizing for a single product recovery system with variable setup numbers, *Eur. J. Oper. Res.* 203 (2010) 326–335, <https://doi.org/10.1016/j.ejor.2009.07.018>.
- [45] M.A. Ilgin, S.M. Gupta, Environmentally conscious manufacturing and product recovery (ECMPRO): a review of the state of the art, *J. Environ. Manage.* 91 (2010) 563–591, <https://doi.org/10.1016/j.jenvman.2009.09.037>.
- [46] A.M.A. El Saadany, M.Y. Jaber, A production/remanufacture model with returns' subassemblies managed differently, *Int. J. Prod. Econ.* 133 (2011) 119–126, <https://doi.org/10.1016/j.ijpe.2010.08.014>.
- [47] M.Y. Jaber, A.M.A. El Saadany, An economic production and remanufacturing model with learning effects, *Int. J. Prod. Econ.* 131 (2011) 115–127, <https://doi.org/10.1016/j.ijpe.2009.04.019>.
- [48] M.Y. Jaber, A.M.A. El Saadany, M.A. Rosen, Simple price-driven reverse logistics system with entropy and exergy costs, *Int. J. Exergy* 9 (2011) 486–502, <https://doi.org/10.1504/IJEX.2011.043921>.
- [49] P. Hasanov, M.Y. Jaber, S. Zolfaghari, Production, remanufacturing and waste disposal models for the cases of pure and partial backordering, *Appl. Math. Model.* 36 (2012) 5249–5261, <https://doi.org/10.1016/J.APM.2011.11.066>.
- [50] A.M.A. El Saadany, M.Y. Jaber, M. Bonney, How many times to remanufacture? *Int. J. Prod. Econ.* 143 (2013) 598–604, <https://doi.org/10.1016/j.ijpe.2011.11.017>.
- [51] N. Matar, M.Y. Jaber, C. Searcy, A reverse logistics inventory model for plastic bottles, *Int. J. Logist. Manag.* 25 (2014) 315–333, <https://doi.org/10.1108/IJLM-12-2012-0138>.
- [52] E. Bazan, M.Y. Jaber, S. Zanoni, A review of mathematical inventory models for reverse logistics and the future of its modeling: an environmental perspective, *Appl. Math. Model.* 40 (2016) 4151–4178, <https://doi.org/10.1016/j.apm.2015.11.027>.
- [53] H. Sun, W. Chen, B. Liu, X. Chen, Economic lot scheduling problem in a remanufacturing system with returns at different quality grades, *J. Clean. Prod.* 170 (2018) 559–569, <https://doi.org/10.1016/j.jclepro.2017.09.184>.
- [54] F.T.S. Chan, N. Li, S.H. Chung, M. Saadat, Management of sustainable manufacturing systems—a review on mathematical problems, *Int. J. Prod. Res.* 55 (2017) 1210–1225, <https://doi.org/10.1080/00207543.2016.1229067>.
- [55] S.E. Marshall, T.W. Archibald, Lot-sizing for a product recovery system with quality-dependent recovery channels, *Comput. Ind. Eng.* 123 (2018) 134–147, <https://doi.org/10.1016/J.CIE.2018.06.004>.
- [56] C.C. Hsieh, H.H. Lai, N.A. Masruroh, Production decision making considering dual material types and setup time uncertainty, *Appl. Math. Model.* 96 (2021) 751–765, <https://doi.org/10.1016/J.APM.2021.02.010>.
- [57] P. Piñeyro, O. Viera, The economic lot-sizing problem with remanufacturing and heterogeneous returns: formulations, analysis and algorithms, *Int. J. Prod. Res.* 60 (2021) 3521–3533, <https://doi.org/10.1080/00207543.2021.1925771>.
- [58] J. Hatchett, P. McNall, D. Schrady, A repairable item inventory model, 1966. <https://calhoun.nps.edu/handle/10945/9625> (accessed February 4, 2022).
- [59] R. Teunter, E. Van der Laan, K. Inderfurth, How to set the holding cost rates in average cost inventory models with reverse logistics? *Omega (Westport)* 28 (2000) 409–415, [https://doi.org/10.1016/S0305-0483\(99\)00070-5](https://doi.org/10.1016/S0305-0483(99)00070-5).
- [60] M.Y. Jaber, R.Y. Nuwayhid, M.A. Rosen, Price-driven economic order systems from a thermodynamic point of view, *Int. J. Prod. Res.* 42 (2004) 5167–5184, <https://doi.org/10.1080/00207540412331281971>.
- [61] J. Vörös, Product balancing under conditions of quality inflation, cost pressures and growth strategies, *Eur. J. Oper. Res.* 141 (2002) 153–166, [https://doi.org/10.1016/S0377-2217\(01\)00272-7](https://doi.org/10.1016/S0377-2217(01)00272-7).
- [62] M.Y. Jaber, M. Bonney, Economic manufacture/order quantity (EMQ/EOQ) and the learning curve: past, present, and future, *Int. J. Prod. Econ.* 59 (1999) 93–102, [https://doi.org/10.1016/S0925-5273\(98\)00027-9](https://doi.org/10.1016/S0925-5273(98)00027-9).
- [63] M.Y. Jaber, J. Peltokorpi, T.L. Smunt, The lot size problem and the learning curve: a review of mathematical modeling (1950's –2020), *Appl. Math. Model.* 105 (2022) 832–859, <https://doi.org/10.1016/J.APM.2022.01.007>.
- [64] S.R. Singh, S. Sharma, M. Kumar, Inventory models with multiple production and remanufacturing batches under shortages, *Control Cybern* 45 (2016) 385–416. https://yadda.icm.edu.pl/baztech/element/bwmeta1.element.baztech-84d23a17-e4cf-415c-828e-52410b02ed1f/c/Singh-Inventory_3_2016.pdf (accessed February 9, 2022).
- [65] <https://www.connectedpapers.com>, (n.d.).
- [66] D.I. Cho, M. Parlar, A survey of maintenance models for multi-unit systems, *Eur. J. Oper. Res.* 51 (1991) 1–23, [https://doi.org/10.1016/0377-2217\(91\)90141-H](https://doi.org/10.1016/0377-2217(91)90141-H).
- [67] P. Hasanov, M.Y. Jaber, N. Tahirov, Four-level closed loop supply chain with remanufacturing, *Appl. Math. Model.* 66 (2019) 141–155, <https://doi.org/10.1016/J.APM.2018.08.036>.
- [68] E. Bazan, M.Y. Jaber, A.M.A. El Saadany, Carbon emissions and energy effects on manufacturing-remanufacturing inventory models, *Comput. Ind. Eng.* 88 (2015) 307–316, <https://doi.org/10.1016/j.cie.2015.07.002>.
- [69] N. Tahirov, P. Hasanov, M.Y. Jaber, Optimization of closed-loop supply chain of multi-items with returned subassemblies, *Int. J. Prod. Econ.* 174 (2016) 1–10, <https://doi.org/10.1016/J.IJPE.2016.01.004>.
- [70] P. Hasanov, M.Y. Jaber, S. Zanoni, L.E. Zavanella, Closed-loop supply chain system with energy, transportation and waste disposal costs, *Int. J. Sustain. Eng.* 6 (2013) 352–358, <https://doi.org/10.1080/19397038.2012.762433>.
- [71] Z. Chen, B. Bidanda, Sustainable manufacturing production-inventory decision of multiple factories with JIT logistics, component recovery and emission control, *Transp. Res. Part E Logist. Transp. Rev.* 128 (2019) 356–383, <https://doi.org/10.1016/J.TRE.2019.06.013>.
- [72] H. Liao, D. Wu, Y. Wang, Z. Lyu, H. Sun, Y. Nie, H. He, Impacts of carbon trading mechanism on closed-loop supply chain: a case study of stringer pallet remanufacturing, *Socioecon. Plann. Sci.* 81 (2022) 101209, <https://doi.org/10.1016/J.SEPS.2021.101209>.
- [73] L.D. Condeixa, P. Silva, D. Moah, B. Farias, A. Leiras, Evaluating cost impacts on reverse logistics using an economic order quantity (EOQ) model with environmental and social considerations, *Cent. Eur. J. Oper. Res.* 30 (2022) 921–940, <https://doi.org/10.1007/S10100-020-00707-4/FIGURES/4>.
- [74] J.D. Abbey, H.N. Geismar, G.C. Souza, Improving remanufacturing core recovery and profitability through seeding, *Prod. Oper. Manag.* 28 (2019) 610–627, <https://doi.org/10.1111/POMS.12937>.
- [75] S. Wei, O. Tang, E. Sundin, Core (product) acquisition management for remanufacturing: a review, *J. Remanuf.* 5 (2015) 1–27, <https://doi.org/10.1186/S13243-015-0014-7>.
- [76] S. Wei, O. Tang, W. Liu, Refund policies for cores with quality variation in OEM remanufacturing, *Int. J. Prod. Econ.* 170 (2015) 629–640, <https://doi.org/10.1016/J.IJPE.2014.12.006>.
- [77] A.L. Guiffreda, M.Y. Jaber, Managerial and economic impacts of reducing delivery variance in the supply chain, *Appl. Math. Model.* 32 (2008) 2149–2161, <https://doi.org/10.1016/J.APM.2007.07.006>.
- [78] M. Porter, J. Heppelmann, How smart, connected products are transforming companies, *Harv. Bus. Rev.* 93 (2015) 96–114.